

Mahmoud Zeidan, PhD



📍 Faculty of Computer and Information Sciences (FCIS)
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Executive Summary

Computer Graphics Scientist with strong expertise in scientific visualization, GPU programming, real-time rendering, and parallel computing. Proven experience designing high-performance software for compute-intensive, GPU-accelerated applications. Skilled in architecting scalable enterprise-grade systems and proficient in C++, Python, CUDA, OpenGL, and Vulkan. Experienced in building visualization systems, geometric processing pipelines, and distributed rendering architectures, with applied machine learning and deep learning to enhance graphics and computational performance.

Research Interests

Computer graphics, scientific visualization, GPU computing, real-time rendering, parallel processing, machine learning, and deep learning.

Education

Feb 2016 – Dec 2025	PhD in Computer Science Karlsruhe Institute of Technology (KIT), Institute for Visualization and Data Analysis Thesis: <i>Glyph-based Display and Level-of-Detail Particle Rendering for Interactive Scientific Visualization</i> Supervisor: Prof. Dr.-Ing. Carsten Dachsbacher	Karlsruhe, Germany
Jun 2009 – Jul 2011	MSc in Computer Science Ain Shams University Thesis: Applying Parallel Processing Approach for Interactive Global Illumination Supervisors: Prof. Taymour Nazmy, Prof. Mohamed Hashem, Dr. Haytham El-Messiry	Cairo, Egypt
Sep 2000 – Jun 2004	BSc in Computer Science, Excellent with Honors Ain Shams University Thesis: "Photonix" – A 3D Modeling Tool Using Photon Mapping	Cairo, Egypt

Experience

- Jan 2026 – Present
Research Scientist and Lecturer, *Faculty of Computer and Information Sciences, Ain Shams University*
Cairo, Egypt
- Conduct research developing deep learning algorithms to enhance scientific visualization, real-time rendering, and parallel computing.
 - Proposal applications for funding acquisition and publishing research findings in reputable journals and international conferences.
 - Teaching Computer Graphics, Spring 2026.
 - Supervise students applying machine learning and deep learning techniques to graphics and visualization problems.
- Aug 2016 – Dec 2025
Research and Teaching Assistant, *Karlsruhe Institute of Technology (KIT)*
Karlsruhe, Germany
- Developed visualization software and GPU-based frameworks for scientific data analysis.
 - Designed and implemented real-time rendering and parallel computing pipelines for large-scale scientific datasets using CUDA and modern C++.
 - Built scalable, high-performance visualization tools for interactive exploration of complex, data-intensive workloads.
 - Taught GPGPU, GPU Computing, Graphics Programming, and Game Development, including lab sessions, assignments, and student project supervision.
- Dec 2005 – Feb 2016
Research and Teaching Assistant, *Faculty of Computer and Information Sciences, Ain Shams University*
Cairo, Egypt
- Conducted research on path tracing and photon mapping for physically based, realistic image synthesis.
 - Implemented and optimized GPU rendering algorithms to accelerate global illumination and improve rendering performance.
 - Developed and evaluated rendering techniques for high-quality image synthesis, bridging computer graphics theory with practical implementation.
 - Taught Mathematics, Physics, Probability & Statistics, and Structured Programming, including tutorials, lab sessions, and student support.
- Oct 2012 – Mar 2013
Research Assistant (part-time), *Microsoft Research ATLC*
Cairo, Egypt
- Developed computer vision tools for object detection and image-based search.
 - Implemented and evaluated feature extraction and matching algorithms to improve retrieval accuracy.
- May 2007 – Dec 2007
Software Engineer, *Ractors Inc.*
Cairo, Egypt
- Designed and developed back-end architecture for high-traffic social media applications using PHP, JavaScript, and HTML.
 - Engineered database-driven features and data-access layers to support concurrent multi-user interaction and growing data volumes.
- Nov 2006 – Apr 2007
Software Engineer, *quTIP Software Company*
Cairo, Egypt
- Designed and developed desktop business applications using C#, .NET, and MS SQL Server.
 - Engineered database-backed features and intuitive interfaces, implementing robust data-access layers and business logic for enterprise workflows.

Publications

- under submission
Mahmoud Zeidan. *Large Particle Datasets Visualization on GPUs*. IVD – KIT.

- 2022 Mahmoud Zeidan, Christoph Peters, Tobias Rapp, and Carsten Dachsbacher. *Versatile Geometric Flow Visualization by Controllable Shape and Volumetric Appearance*. Smart Tools and Applications in Graphics (STAG). [\[link\]](#)
- 2020 Mahmoud Zeidan, Tobias Rapp, Christoph Peters, and Carsten Dachsbacher. *Moment-based Opacity Optimization*. Eurographics Symposium on Parallel Graphics and Visualization (EGPGV). [\[link\]](#)
- 2015 Mahmoud Zeidan, Taymour Nazmy, and Mostafa Aref. *GPU-based Out-of-Core HLBVH Construction*. Eurographics Symposium on Rendering (EGSR) – Experimental Ideas & Implementation. [\[link\]](#)

== Technical Strengths

Programming Languages:	C/C++, Python, Matlab – strong OOP design, clean and efficient code, SOLID principles.
GPU & Visualization:	OpenGL, GLSL, Vulkan, CUDA – advanced GPU programming, shader development, and performance tuning.
Database Systems:	Database modeling and design, SQL Server, MySQL, relational schema design.
Engineering Tools:	Git, CMake, Bash – workflow automation, CI, and cross-platform development.
Visualization & Design:	Blender, GIMP, Inkscape, Clip Studio Paint – research visualization and editing.
Operating Systems:	Linux (primary), Windows – system configuration and cross-platform workflows.

== Languages

Arabic : Native
 English : Professional working proficiency
 German : Proficient in daily communication

== Hobbies and Personal Interests

Painting and modeling on physical and digital media, woodworking, nature walks, and writing and blogging in Arabic.

== References

- KIT – IVD **Prof. Dr.-Ing. Carsten Dachsbacher**
 Head of Computer Graphics Group, Institute for Visualization and Data Analysis (IVD)
 Karlsruhe Institute of Technology (KIT), Adenauerring 2, 76131 Karlsruhe, Germany
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- TU Delft **Dr. Christoph Peters**
 Computer Graphics and Visualization Group, Delft University of Technology (TU Delft)
 Van Mourik Broekmanweg 6, 2628 XE Delft, The Netherlands
 ✉ c.j.p.peters@tudelft.nl
- KIT – IAI **Mahmoud Salem, M.Sc.**
 Machine Learning and Robotics Scientist, Institute for Automation and Applied Informatics (IAI)
 Karlsruhe Institute of Technology (KIT), Hermann-von-Helmholtz-Platz 1, 76344 Eggenstein-Leopoldshafen, Germany
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